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ORIGINAL ARTICLE

Stock movement prediction with multimodal stable fusion via gated cross-attention mechanism

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Abstract

The accurate prediction of stock movements is crucial for investment strategies. Stock prices are subject to the influence of various forms of information, including financial indicators, sentiment analysis, news documents, and relational structures. Predominant analytical approaches, however, tend to address only unimodal or bimodal sources, neglecting the complexity of multimodal data. Further complicating the landscape are the issues of data sparsity and semantic conflicts between these modalities, which are frequently overlooked by current models, leading to unstable performance and limiting practical applicability. To address these shortcomings, this study introduces a novel architecture, named Multimodal Stable Fusion with Gated Cross-Attention (MSGCA), designed to robustly integrate multimodal input for stock movement prediction. The MSGCA framework consists of three integral components: (1) a trimodal encoding module, responsible for processing indicator sequences, dynamic documents, and a relational graph, and standardizing their feature representations; (2) a cross-feature fusion module, where primary and consistent features guide the multimodal fusion of the three modalities via a pair of gated cross-attention networks; and (3) a prediction module, which refines the fused features through temporal and dimensional reduction to execute precise movement forecasting. Empirical evaluations demonstrate that the MSGCA framework exceeds current leading methods, achieving performance gains of 8.1%, 6.1%, 21.7% and 31.6% on four multimodal datasets, respectively, attributed to its enhanced multimodal fusion stability.

Keywords Stock movement prediction · Multimodal fusion · Gated cross-attention

دمج ميزات متعدد الوسائط للتنبؤ بحركة الأسهم
آلية الانتباه المتقاطع البوابي

Introduction

The increasing capitalization of global stock markets and the rapid growth in data availability have motivated extensive research into stock movement prediction. Accurately forecasting stock trends is critical for investment decision-making and risk management. Traditional approaches often rely on singular data sources such as financial indicators [2, 3], social media sentiment [4, 5], or market relationship networks [1, 6]. However, these methods overlook the potential benefits of integrating heterogeneous information, which

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may better capture the multifaceted factors influencing stock prices.

In real-world financial markets, stock prices are affected by a variety of signals including numerical indicators (e.g., open, close, and high prices), textual documents (e.g., news reports and tweets), and relational structures (e.g., industry and investment relationships). The incorporation of such multimodal data can provide a more comprehensive understanding of market dynamics. Nonetheless, existing multimodal models typically address only unimodal or bimodal sources [7–10] and often employ simple fusion strategies, such as vector concatenation or basic attention mechanisms, which fail to resolve challenges like data sparsity and semantic conflicts. These issues are clearly illustrated in Fig. 1, where the empirical case from China's SSE STAR Market (Sci-Tech Innovation Board) shows significant gaps in news document coverage during holidays and a marked semantic conflict between the reported news and the observed price trajectories (the red markers in Fig. 1 emphasize these discrepancies).

Motivated by these challenges, this study poses the following research questions: Is it possible to enhance multimodal stock movement prediction for fine-grained trend forecasting by employing a stable fusion framework? How can the integration of a primary modality guide the fusion of auxiliary modalities to mitigate the impact of noisy or conflicting data? To address these questions, we propose a novel framework called Multimodal Stable Fusion via Gated Cross-Attention (MSGCA). This framework is designed to integrate three key modalities–indicator sequences, dynamic documents, and relational graphs–to predict stock trends at a fine-grained level (up, flat, and down).

The proposed method follows an encoder-fusion-decoder architecture. In the encoding phase, a trimodal encoder is deployed: numerical indicators are mapped via linear projections, dynamic documents are transformed into latent representations using a pre-trained large language model (text-embedding-ada-002), and relational graphs are encoded with a graph attention network (GAT). In the fusion phase, a gated cross-attention mechanism is introduced to progressively integrate the encoded features. The stable and complete information in the indicator sequences is employed as the primary guide to fuse auxiliary modalities, thereby alleviating issues of data sparsity and semantic conflicts. Finally, in the decoding phase, two multilayer perceptron (MLP) networks aggregate the temporal and feature dimensions to produce fine-grained movement probabilities.

Our contributions are summarized as follows:

1. We propose MSGCA, a novel framework for fine-grained stock movement prediction that effectively integrates indicator sequences, dynamic documents, and relational graphs.
2. We design a gated cross-attention mechanism that leverages the stability of financial indicators to guide the fusion of multimodal features, addressing the challenges of noisy and conflicting information.
3. We introduce a new dataset, InnoStock, which compiles financial news, stock prices, and sector relationships for technology companies listed on Sci-Tech Innovation Board, thereby enriching the resources available for research in this area.
4. Extensive experiments on four multimodal datasets demonstrate that MSGCA outperforms state-of-the-art methods in terms of prediction accuracy and computational efficiency.

The remainder of the paper is organized as follows. First, we review the related work and make a brief comparison between our method and previous studies in Sect. "Related work". Then we provide the preliminary definitions and task formulation in Sect. "Definition and formulation". The implementation of our MSGCA stock movement prediction method is presented in Sect. "Methodology". Sect. "Experiments" shows the experiments and analysis, followed by the conclusion and future work in Sect. "Conclusion and future work".

Related work

Stock movement prediction

Stock movement prediction is challenging due to the market being highly stochastic and temporally dependent on predictions from chaotic data [7]. We summarize the existing stock movement prediction methods in Table 1. Traditional methods focus on utilizing single-modal input with statistical [4, 17] or simple deep learning methods [1–3]. For example, an early work named JST [4] extracts topics and sentiments simultaneously using a linear kernel SVM and utilizes them for stock market prediction. MFNN [3] learns an end-to-end model with integration of convolutional and recurrent neurons for feature extraction on financial time series samples. TGC [1] proposes a temporal graph convolution, which jointly models the temporal evolution and the relation network of stocks. However, all of the above methods focus on a singular modality in one of indicators, text, or graph, thereby providing limited information for the multifaceted task. In addition, these works utilize LSTM or CNN variants that are limited in their representational capabilities. To address these problems, recent efforts on stock movement prediction focus on introducing more types of information [7, 8, 29] and using modern deep learning methods such as the attention mechanism [6, 15, 26, 35–37] and large language models [11–13, 18]. For example, TRAN [6] proposes a time-aware relational

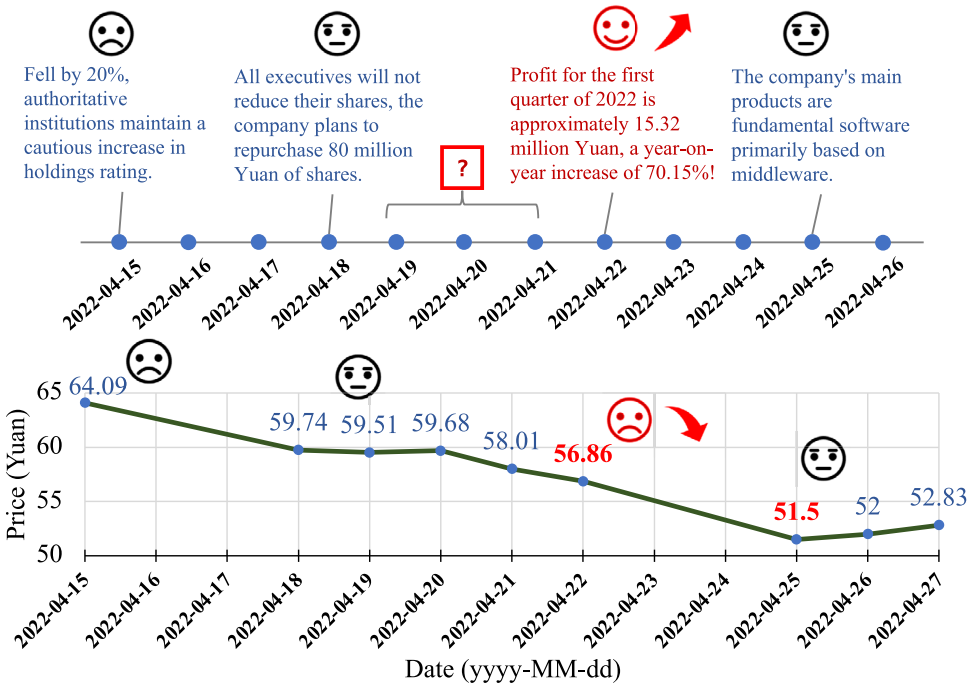


Fig. 1 An empirical case from China's SSE STAR Market illustrates the challenges associated with the dataset of multimodal stock information. First, the dataset exhibits gaps in the news documents that align with price indicators, barring holiday periods. Additionally, a discernible semantic conflict is observed between the reported news content and concurrent price trajectories. These discrepancies are marked in red for emphasis

attention network for graph-based stock movement prediction to capture time-varying correlation strengths between stocks. MAC [8] proposes a multisource aggregated method that incorporates the numerical features, market-driven news, and the sentiments of their related stocks. ALSTM [36] complete the task using components that include feature mapping, LSTM, temporal attention, and prediction layer. SLOT [15] learns the unified embeddings of prices and tweets from self-supervised learning and uses ALSTM for prediction. Recent approaches such as LLMFactor [66] and FinedFMP [67] have introduced large language models to enhance textual feature representation for stock trend analysis. LLMFactor fine-tunes LLMs with factor-based supervision, while FinedFMP integrates financial news with expert knowledge for forecasting.

All the aforementioned methods focus on predicting the up and down trends of stock prices given various stock features. However, the flat trend of a stock is also important for prediction, which indicates that the forces of demand are nearly equal before a new period. Some previous studies aim to solve a fine-grained prediction but with single-modal input. In contrast, our proposed MSGCA tries to perform a stable multimodal fusion by introducing three types of modalities (indicator, documents, and graph) to solve the task with labels of up, flat and down, which fills the gap for fine-grained prediction with multimodal features as shown in Table 1.

Multimodal fusion strategies

Multimodal fusion is the process of combining collected data from various modalities for analysis tasks [39], and the fusion of multimodal data can provide surplus information with an increase in accuracy of the overall result [40]. Most early multimodal fusion strategies implement the framework of encoder and concatenation process, which focuses on using a vectorized concatenation operator to generate the final features [21–23]. These simple fusion methods overlook the interference of noisy information while performing useful feature extraction [41]. Some other works utilize the attention and co-attention mechanism to simultaneously integrate information from all modalities [24, 42, 43], but ignore the differences between modalities for the targeted tasks. These methods involve on filtering noise while jointly learning the multimodal features. Modern approaches focus on more sufficient multimodal interactions [25, 44]. These studies employ self-attention and co-attention for intro-modal and inter-modal interaction, respectively, but suffer from using redundant information and architecture during the process, and further impair the efficiency of the framework [45].

Considering the problem of noise interference, modality differences, and efficiency degeneration, our MSGCA focus on the stock movement prediction task, which achieves a stable multimodal fusion using a gated cross-attention mechanism guided by the main modality and features step-by-step during the fusion process.

Table 1 Summarization of existing stock movement prediction methods. Our MSGCA is highlighted in bold

Input Type Methods Task Type	Unimodal features	Multimodal features
Coarse-Grained Prediction	JST [4]	TRAN [6]
	CapTE [5]	StockNet [7]
	MFNN [3]	MAC [8]
	ALSTM [36]	AStock [18]
	DTML [35]	ESTIMATE [26]
		SLOT [15]
Fine-Grained Prediction	Xuanwu [17]	**MSGCA**
	T-LSTM [16]	
	TASA [38]	

Our proposed MSGCA covers both multimodal features and fine-grained prediction settings

Cross and gated-attention mechanisms

A powerful and robust model to represent features from multiple input sequences requires components that can effectively aggregate information relevant to the task from each source, and cross-attention is considered as an effective mechanism for feature integration [46]. Existing cross-attention studies treat one sequence as query, and the other sequences as key and value to perform the encoder [47–50]. For example, IRENE [49] fuses textual and visual features with two bidirectional cross-attention blocks to achieve mutual integration between two types of input. Another work [46] implements a tri-modal cross-attention architecture to perform cross-attention with combinatorial input from three sources. In addition, the gating mechanism has been shown to be essential for recurrent neural networks (RNNs) to achieve state-of-the-art performance [51]. Gated linear unit (GLU) can reduce the problem of gradient vanishing by providing a linear path for gradients while maintaining non-linear capabilities [52]. Gated attention unit to formulate attention and GLU as a unified layer, which allows the use of a much simpler attention mechanism and results in higher computational efficiency [53].

Following the above studies, our MSGCA employs cross-attention and gated-attention mechanisms to solve the feature integration problem from multiple sources to predict stock movement. Meanwhile, We distinguish stock-related information into primary and auxiliary features, and employ a fusion method involving three modalities through two consecutive gated cross-attention mechanisms to maintain high efficiency.

Definition and formulation

In this section, we provide some definitions and formulation of our task to better understand our method in this study.

Definition 1 ***Multimodal source.*** The multimodal source $\mathcal{M}$ of a stock consists of various types of information as input to our stock movement prediction framework. In our study, we focus on three types of sources for each stock, represented as $\mathcal{M} = \{\mathcal{I}, \mathcal{D}, \mathcal{G}\}$, and each modality is formulated as follows.

- **Indicator sequence**, expressed as $\mathcal{I} = \{i_1, i_2, ..., i_t\}$, where i_t is the numerical indicator, such as the open price, close price, or highest price of a stock produced at timestamp t.
- **Dynamic document**, defined as timestamped unstructured textual content associated with the stock (e.g. news, social media posts, analyst reports), represented as $\mathcal{D} = \{d_1, d_2, ..., d_t\}$ and $d_t = \{p_t^1, p_t^2, ..., p_t^n\}$, where d_t is the set of tweets or news in the published time stamp t, and p_t^n is the n-th paragraph that composes d_t.
- **Relational graph**, defined as any structured representation of inter-entity relationships among stocks, industries, or other economic factors [37, 54], denoted as $\mathcal{G} = \{\mathcal{E}, \mathcal{R}, \mathcal{U}\}$, where $\mathcal{E}$ is the set of entities and $\mathcal{R}$ is the set of relations. $\mathcal{U} = \{(e_s, r, e_d)|e_s, e_d \in \mathcal{E}, r \in \mathcal{R}\}$ is the set of triplets in the graph $\mathcal{G}$, where e_s, e_d and r are the start entity, the end entity, and the relation, respectively.

According to the data we collect from real-world sources, the indicator sequence and dynamic documents of a stock contain a time dimension. Whereas, the relational graph is static.

Definition 2 ***Multimodal fused feature.*** The multimodal fused feature $\boldsymbol{x}$ of a stock is transformed from its multimodal sources $\mathcal{M}$ by an encoder-fusion process, denoted $\boldsymbol{x} = \boldsymbol{v}_i \circ \boldsymbol{v}_d \circ \boldsymbol{v}_g$, where $\circ$ is the multimodal fusion method. The indicator feature $\boldsymbol{v}_i$, the document feature $\boldsymbol{v}_d$, and the graph feature $\boldsymbol{v}_g$ are learned from their corresponding encoders.

Definition 3 ***Fine-grained movement prediction.*** Fine-grained movement prediction is to generate the probability for

each trend label l from the set of labels L using the fused multimodal feature of a stock $\boldsymbol{x}$, denoted $Pr(y \in L|\boldsymbol{x}) = f(\boldsymbol{x})$, where f is the prediction function, and $L = \{up, flat, down\}$.

Objective: This study aims to predict the trend of price movement of a stock in the timestamp $t+1$, given the multimodal source of this stock from timestamp 1 to timestamp t. The problem formulation is given as:

- The input is the multimodal source of a stock $\mathcal{M} = \{\mathcal{I}, \mathcal{D}, \mathcal{G}\}$ which contains an indicator sequence $\mathcal{I}$, a dynamic document $\mathcal{D}$ and a static graph $\mathcal{G}$.
- The output is a probability distribution $p = Pr(y \in L|\boldsymbol{x})$ for the set of target trend labels L generated by an encoder-fusion-decoder process.

Methodology

Framework of MSGCA

The framework of our proposed method **MSGCA** is shown in Fig. 2. The procedure is structured into three phases, adhering to an encoder-fusion-decoder architecture. During the multimodal encoding phase, projections from three modalities are transposed into a cohesive latent space through employing various representational methods as encoders. The gated cross-feature fusion phase implements a stable multimodal integration strategy to synthesize a unified feature representation through a pair of gated cross-attention networks. The movement prediction phase compresses both temporal and feature dimensions, aligning them with the granularity of the trend labels, thereby enabling precise prediction of stock movement at timestamp $T+1$. In Table 2, we summarize the notations used in our framework.

Multimodal source encoding

Analysis of stock movement is based on the idea that historical market behavior can inform traders about potential future trends. Conventional methods for stock movement prediction focus on employing statistical models on a range of financial indicators, including open prices, highest prices, lowest prices, closed prices, and trading volumes [2, 4, 17]. Recent deep learning approaches are apt to introduce more types of data sources, such as tweet text, and market relational data [6, 15, 37]. The above methods encompass the use of indicators, documents, and graphs, which inspires us to concentrate our efforts on the management of these three distinct data sources. We implement a trimodal encoding module as the first phase of MSGCA to concurrently encode each type of multimodal information into a unified latent dimensional space. Each encoder is described as follows.

1) Indicator sequence encoding: In our framework, the indicator sequence of a stock is considered its primary modality, which is characterized by data completeness and continuity. MSGCA applies three fundamental indicators, including closed prices, open prices, and highest prices, to demonstrate its proficiency in processing sequences of multiple indicators. Each indicator category is mapped into a d-dimensional latent feature space using a specified linear layer. Then, all three indicator features are concatenated and transformed into a unified feature sequence to facilitate the upcoming multimodal fusion. The encoding process for indicator sequences is formulated as the following equations:

$$\mathbf{v}_i = (\mathbf{v}_c \oplus \mathbf{v}_o \oplus \mathbf{v}_h)\mathbf{W}_i + b_i \tag{1}$$

$$\mathbf{v}_c = \mathbf{s}_c\mathbf{W}_{ic} + b_{ic} \tag{2}$$

$$\mathbf{v}_o = \mathbf{s}_o\mathbf{W}_{io} + b_{io} \tag{3}$$

$$\mathbf{v}_h = \mathbf{s}_h\mathbf{W}_{ih} + b_{ih} \tag{4}$$

where $\mathbf{s}_c$, $\mathbf{s}_o$, and $\mathbf{s}_h \in \mathbb{R}^{t\times 1}$ are indicator sequences of closing, open, and highest prices, respectively, with time dimension t, $\mathbf{W}_{ic} \in \mathbb{R}^{d\times d}$, $\mathbf{W}_{io} \in \mathbb{R}^{d\times d}$, and $\mathbf{W}_{ih} \in \mathbb{R}^{d\times d}$ are learnable weights of these corresponding indicators. $\mathbf{v}_c$, $\mathbf{v}_o$, and $\mathbf{v}_h \in \mathbb{R}^{t\times d}$ are the latent embeddings of these indicators, $\oplus$ is the concatenation operator, $\mathbf{W}_i \in \mathbb{R}^{3d\times d}$ is the learnable weight of the linear projection, $\mathbf{v}_i \in \mathbb{R}^{t\times d}$ is the overall feature of an indicator sequence.

2) Dynamic document encoding: Social media content and news reports on a stock can reveal investor sentiment and corporate developments, which are factors to influence future stock valuations. These sources of textual information are presented as dynamic documents that require feature extraction for our predictive analysis. However, due to the scarcity of data acquisition channels, documents about a particular stock do not invariably accompany its daily market activity. Moreover, Large Language Models (LLMs) such as ChatGPT[1] and LLaMA [55] have excelled in various natural language tasks, which inspires us to employ a pretrained LLM to perform the document encoding step. In our framework, we use text-embedding-ada-002[2] from OpenAI as our parameter-fixed LLM to convert textual documents into high-dimensional latent embeddings. Those vectorized representations are then subjected to a dimensionality alignment with the indicator modality through a linear layer. While the LLM encoder provides semantically rich representations, it is sensitive to input variability, hence, preprocessing is crucial for consistent results. Before generating embeddings, we preprocess the textual content by lowercasing and removing

[1] https://chat.openai.com/

[2] https://platform.openai.com/docs/guides/embeddings.

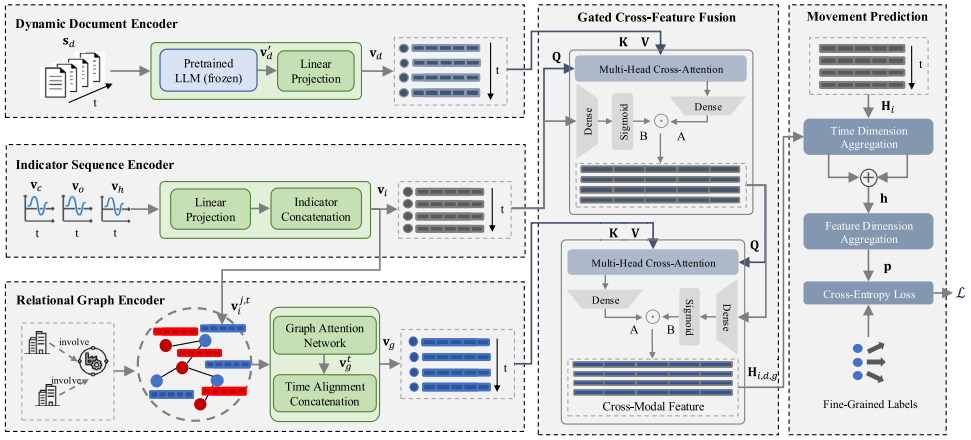


Fig. 2 Framework of MSGCA. Indicator sequences, dynamic documents, and a relational graph are encoded with a linear projection, a pre-trained large language model (LLM), and a graph attention network (GAT), respectively. The framework then fuses indicator, document, and graph features in two gated cross-attention steps. First, indicator features guide the integration of dynamic documents via multi-head cross-attention and a gating unit. Second, the intermediate fused result serves as the query to integrate graph features through another gated cross-attention step. This design ensures robust fusion by progressively stabilizing multimodal feature integration. Finally, the movement predictor transforms the fused features into the probabilities for each trend label by performing aggregations along the temporal and feature dimensions with two MLP networks. All parameters in this framework are optimized by backpropagation with the overall loss $\mathcal{L}$

Table 2 Notations used in our method. Vectors and tensors are denoted in bold ‘

Notation	Meaning
$\mathbf{s}_c, \mathbf{s}_o, \mathbf{s}_h$	indicator sequences of closed, open, and highest price
$\mathbf{s}_d$	input sequence of dynamic documents
$\mathbf{v}_i, \mathbf{v}_d, \mathbf{v}_g$	latent representations of three modalities
$\mathcal{F}(\cdot), \mathcal{T}(\cdot)$	functions of filling zero vectors and getting text embeddings
$\mathbf{v}_d^{'}$	document embedding generated from an LLM
$\mathbf{W}, b, b^{'}$	model weights and bias to be learned
$\mathbf{H}, \mathbf{h}$	updated stock representation after feature fusion
$\mathbf{Q}, \mathbf{K}, \mathbf{V}$	query, key, and value for cross-attention layers
K, M	number of heads for multi-head GAT and cross-attention
$d, d^{'}$	embedding dimensions of learnable parameters and features
$\mathbf{p}, \mathbf{l}$	multi-class probability output and label vectors
$\mathcal{L}$	overall loss of the framework

punctuation and stopwords. We rely on the LLM's inherent ability to handle semantic ambiguity and ensure consistent representation of financial entities without explicit NER. Furthermore, to address instances where document timestamps are missing, we populate the temporal sequence with zero vectors to maintain consistency with the corresponding indicator features. The encoding process for dynamic documents is described by the following equations:

$$\mathbf{v}_d = \mathcal{F}(\mathbf{v}_d^{'}\mathbf{W}_d + b_d) \tag{5}$$

$$\mathbf{v}_d^{'} = \mathcal{T}(\mathbf{s}_d, LLM) \tag{6}$$

where $\mathcal{T}(\cdot)$ is the function to get the latent text embeddings $\mathbf{v}_d^{'} \in \mathbb{R}^{t^{'} \times 1536}$ from the given document sequence $\mathbf{s}_d$ by calling the API service provided by OpenAI, using a pre-trained large language model LLM, and $\mathbf{W}_d \in \mathbb{R}^{1536 \times d}$ is the learnable weight of the linear projection. The dynamic document feature $\mathbf{v}_d \in \mathbb{R}^{t \times d}$ is aligned to its respective indicator sequence through a zero vector filling function $\mathcal{F}(\cdot)$.

3) Relational graph encoding: Relationships between stocks often reflect sectoral and financial influences, which are useful in predicting stock price movement. We collect data describing the interconnections between stocks and their respective sectors and construct a relational graph to serve as our graph modality. As industry-specific fluctuations can concurrently affect stocks within the same sector, we employ a graph attention network (GAT) [56] as the graph encoder to capture these correlated variations. Meanwhile, to make the information related to each stock propagate through the

graph, nodes representing individual stocks are initialized with vectors corresponding to indicator features at discrete time intervals. This operation facilitates the extension of the temporal dimension within the graph modality, ensuring alignment with the other two modalities. The relational graph encoding process is denoted by the following equations:

$$\mathbf{v}_g = [\mathbf{v}_g^1, \mathbf{v}_g^2, ..., \mathbf{v}_g^t] \tag{7}$$

$$\mathbf{v}_g^t = \sigma\left(\frac{1}{K}\sum_{k=1}^{K}\sum_{j\in\mathcal{N}}\alpha_j^k \mathbf{h}_j^t \mathbf{W}^k\right) \tag{8}$$

$$\mathbf{h}_j^{t(0)} = \mathbf{v}_i^{j,t} \tag{9}$$

where the overall graph embedding $\mathbf{v}_g \in \mathbb{R}^{t\times d}$ is composed of embeddings from each timestamp $\mathbf{v}_g^t \in \mathbb{R}^{1\times d}$. K is the number of heads to perform multi-head attention. $\mathcal{N}$ is the neighborhood of the central node. α_j^k is the attention coefficient for each neighboring node in each head. $\mathbf{W}^k \in \mathbb{R}^{d\times d}$ is the learnable weight matrix, and $\mathbf{h}_j^t \in \mathbb{R}^{1\times d}$ is the hidden embedding of node j in timestamp t from the previous layer of GAT. σ is an activity function. The initial embedding $\mathbf{h}_j^{t(0)}$ of the node j in timestamp t is its indicator feature $\mathbf{v}_i^{j,t} \in \mathbb{R}^{1\times d}$.

The indicator, document, and graph encoders output vectors $\mathbf{v}_i$, $\mathbf{v}_d$, and $\mathbf{v}_g$, respectively, which serve as inputs to the gated cross-attention module described in the following fusion stage.

Gated cross-feature fusion

Data sparsity and semantic conflicts within multimodal fusion significantly compromise stock movement prediction efficacy, as described in our secondary challenge. Meanwhile, stock indicator sequences typically exhibit superior data integrity compared to other data modalities and ought to be prioritized as the primary input in our analysis. In the second module of MSGCA, we introduce an innovative gated cross-attention network designed to integrate three distinct modalities. This fusion is orchestrated by leveraging primary and stable features to guide the fusion process, distinguishing our approach from all other existing stock movement prediction methods.

The gated cross-attention network is inspired by studies on efficient transformers [53], cross-attention fusion [47] and gated networks [52]. To integrate features from two modalities, the network initiates by executing a deeply interactive feature fusion via multi-head cross-attention mechanisms, thereby acquiring features in an unstable state. Then, using a stable feature as a discriminator, the network executes a gated layer operation to select the useful information from the unstable features for the subsequent processing stages.

To provide a clearer understanding of our fusion strategy, we elaborate the gated cross-attention mechanism as a two-stage process: In the first stage, the indicator sequence, our primary modality, guides the fusion with dynamic documents via a cross-attention operation, followed by a gating mechanism that filters unstable features. In the second stage, the stable fused output from the first step is used as the guiding signal to incorporate relational graph features via a similar cross-attention and gating process. This progressive design ensures that modality fusion is anchored in stable, semantically aligned information. The details of this procedure are illustrated as follows.

1) Fusion of indicator and document (unstable cross-attention): During the fusion of indicator sequences with dynamic documents, a multi-head cross-attention network is employed to realize the deep integration of the two modalities. We configure the indicator features as queries, and the document features as keys and values, thereby executing a multi-head cross-attention process. This fusion process is described in the following equations:

$$\mathbf{H}_{i,d}^l = softmax\left(\frac{1}{M}\sum_{m=1}^{M}\frac{\mathbf{Q}_m^l(\mathbf{K}_m^l)^\top}{\sqrt{d^{'}}}\right)\mathbf{V}_m^l \tag{10}$$

$$\mathbf{Q}_m^l = \mathbf{H}_i^l\mathbf{W}_m^Q,$$
$$\mathbf{K}_m^l = \mathbf{H}_d^l\mathbf{W}_m^K, \mathbf{V}_m^l = \mathbf{H}_d^l\mathbf{W}_m^V \tag{11}$$

where $\mathbf{H}_{i,d}^l$ is the unstable features from the fusion of indicator sequences and dynamic documents using multi-head cross-attention. M is the number of heads. $d^{'} = M \times d$. $\mathbf{W}_m^Q$, $\mathbf{W}_m^K$, and $\mathbf{W}_m^V \in \mathbb{R}^{d\times d^{'}}$ are learnable weights of the query $\mathbf{H}_i^l$, key $\mathbf{H}_d^l$, and value $\mathbf{H}_d^l$ from the l-th layer.

2) Fusion of indicator and document (stable gated feature selection): The previously integrated features exhibit instability as a result of data sparsity and semantic conflicts, as described in Fig. 1. So a step of latent feature selection is operated using a gated network architecture, which is guided by indicator sequences (our primary modality), in order to extract stable multimodal features. The gated feature selection is formulated as the following equations:

$$\mathbf{H}_{i,d} = \mathbf{H}_a \odot \mathbf{H}_b \tag{12}$$

$$\mathbf{H}_a = \mathbf{H}_{i,d}^l\mathbf{W}_a + b \tag{13}$$

$$\mathbf{H}_b = Sigmoid(\mathbf{H}_i\mathbf{W}_b + b^{'}) \tag{14}$$

where $\mathbf{H}_a$ and $\mathbf{H}_b$ are the unstable-fused feature and main modality feature, respectively. $\mathbf{W}_a \in \mathbb{R}^{t\times d}$ and $\mathbf{W}_b \in \mathbb{R}^{t\times d}$ are learnable weights. $b \in \mathbb{R}^d$ and $b^{'} \in \mathbb{R}^d$ are learnable biases. $\odot$ is the element-wise product operator between matrices. Finally, we get the stable fused feature $\mathbf{H}_{i,d}$ from indicator sequences and dynamic documents.

3) Fusion with graph (expansion to the third modality): Similarly, the fusion of the third modality harnesses the stable characteristic derived from the previous gated cross-attention block to guide the subsequent stage of fusion. To integrate information from the graph modality, a secondary cross-attention network is introduced as the following equations:

$$\mathbf{H}^l_{i,d,g} = softmax\left(\frac{1}{M}\sum_{m=1}^{M}\frac{\mathbf{Q}^l_m(\mathbf{K}^l_m)^\top}{\sqrt{d'}}\right)\mathbf{V}^l_m, \tag{15}$$

$$\mathbf{Q}^l_m = \mathbf{H}^l_{i,d}\mathbf{W}^Q_m, \mathbf{K}^l_m = \mathbf{H}^l_g\mathbf{W}^K_m, \mathbf{V}^l_m = \mathbf{H}^l_g\mathbf{W}^V_m \tag{16}$$

where $\mathbf{H}^l_g$ is the hidden features of the graph modality. $\mathbf{H}^l_{i,d,g}$ is the unstable features which fuse three modalities. Finally, the stable fused feature of a stock derived from three modalities can be computed as the following equations:

$$\mathbf{H}_{i,d,g} = \mathbf{H}_a \odot \mathbf{H}_b, \tag{17}$$

$$\mathbf{H}_a = \mathbf{H}^l_{i,d,g}\mathbf{W}_a + b, \tag{18}$$

$$\mathbf{H}_b = Sigmoid(\mathbf{H}_{i,d}\mathbf{W}_b + c) \tag{19}$$

where $\mathbf{H}_{i,d,g}$ is the final stable feature across the information of three modalities guided by the feature $\mathbf{H}_{i,d}$ from the previous fusion stage. It is worthy to acknowledge that the scalability of our gated cross-attention mechanism extends to the integration of additional modalities within a sequential fusion paradigm.

Fine-grained movement prediction

To predict stock movements, we should convert the integrated features into trend labels. In the third module of MSGCA, we compress the temporal and feature dimensions of stocks with two multi-layer perceptron (MLP) networks to map the fused features into the label space. The detailed steps are described below.

1) Time dimension aggregation: Unlike other sequential methods such as LSTM [57] and self-attention [58] applied to reduce the time dimension when calculating the overall representations of sequences, we directly use an MLP network for efficient dimension reduction. Meanwhile, considering that we only employ stable features as the gate to choose useful information when performing gated cross-attention, we further concatenate the indicator encoding features and the cross-modality features in this stage, to provide a chance to make the prediction directly with the primary features and avoid the potential usage of extreme noisy information from other modalities. This process is formulated as the following equations:

$$\mathbf{h} = \mathbf{h}_{i,d,g} \oplus \mathbf{h}_i \tag{20}$$

$$\mathbf{h}_{i,d,g} = MLP_t(\mathbf{H}_{i,d,g}) \tag{21}$$

$$\mathbf{h}_i = MLP_t(\mathbf{H}_i) \tag{22}$$

where $MLP_t(\cdot)$ is the time aggregation network that gradually reduces the time dimension from t to 1 using three linear layers followed by three activation functions. $\mathbf{H}_{i,d,g}$ and $\mathbf{H}_i$ are cross-modal features and indicator sequence features, respectively. $\oplus$ is the concatenation operator to obtain the features with the time dimension removed $\mathbf{h} \in \mathbb{R}^{2\times d}$.

2) Feature dimension aggregation: Our last step is to project the overall sequence features in the latent space onto each trend label to complete the task. We implement another MLP network to gradually reduce the feature dimension from $2 \times d$ to 3, which is the number of trend categories, to represent the probabilities of each label. This step is illustrated as the following equation:

$$\mathbf{p} = MLP_f(\mathbf{h}) \tag{23}$$

, where $MLP_f(\cdot)$ is the feature aggregation network to gradually reduce feature dimension from d to 3 using the same structure as $MLP_t(\cdot)$. $\mathbf{p} \in \mathbb{R}^3$ is the probability vector with feature dimension reduced from $\mathbf{h}$.

3) Multi-label loss function: With the probability for each label (up, flat, and down) is obtained, a multi-class cross-entropy function is employed to calculate the loss of the overall process, which is formulated as the following equation:

$$\mathcal{L} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{c=1}^{C} log\frac{exp(\mathbf{p}_{n,c})}{\sum_{i=1}^{C} exp(\mathbf{p}_{n,i})}\mathbf{l}_{n,c} \tag{24}$$

where N is the number of batches. $\mathbf{p}_{n,c}$ is the probability of the class label $c \in \{1, 0, -1\}$ for the n-th batch, where label 1, 0, and -1 stands for the trends of up, flat, and down, respectively. $\mathbf{l}_{n,c}$ is the label of the n-th sample and the c-th class from the label vector $\mathbf{l}$. $\mathcal{L}$ is the mean value of the overall loss.

Pseudocode of training MSGCA

To conclude the training process of MSGCA with the above phases, the pseudocode is illustrated as Algorithm 1.

Experiments

We conduct experiments to answer the following research questions on the performance of MSGCA:

RQ1 **Prediction performance (Section V-C):** Does MSGCA outperform previous methods in multimodal stock movement prediction?

Algorithm 1 Pseudocode of training MSGCA

Input: Indicator sequence: $\mathcal{I} = \{i_1, i_2, ..., i_t\}$, Dynamic documents: $\mathcal{D} = \{d_1, d_2, ..., d_t\}$, Relational graph: $\mathcal{G} = \{\mathcal{E}, \mathcal{R}, \mathcal{U}\}$
Arguments:
Hidden dimension: d, Window size: ws, Epoch number: en, Batch size: bs, Learning rate: lr
Output: Training loss: $\mathcal{L}$, Model parameters: $\boldsymbol{\theta}$

```
Get embeddings of dynamic documents v'_d from LLM
Get indicator sequences s_c, s_o, s_h
Split instances according to window size ws
Sample batched training set B according to batch size bs
for epoch ∈ 1, ..., en do
    for batch ∈ B do
        Calculate indicator features v_i using Eq. 1-4
        Calculate document features v_d using Eq. 5-6
        Calculate graph features v_g using Eq. 7-9
        Get fused features H^l_{i,d} from indicator and document modalities using Eq. 10-11
        Get stable features H_{i,d} using Eq. 12-14
        Get fused features H^l_{i,d,g} from previous features and graph modality using Eq. 15-16
        Get stable features H_{i,d,g} using Eq. 17-19
        Get aggregated features h using Eq. 20-22
        Get probability vectors p using Eq. 23
        Calculate multi-label loss L using Eq. 24
        Return L
    end for
    Update θ with backpropagation from loss L using Adam optimizer with learning rate lr
end for
Return model parameter θ
```

RQ2 Ablation study (Section V-D): How do fusion strategies, language encoders, and graph encoders affect the performance of MSGCA?

RQ3 Modality effect (Section V-E): How do different modalities affect the performance of MSGCA? Which modality contributes the most?

RQ4 Stable fusion study (Section V-F): Does the gated cross-attention mechanism in MSGCA work to stabilize multimodal fusion?

RQ5 Hyperparameter effect (Section V-G): How do hidden dimension sizes, time window sizes and learning rates affect MSGCA performance?

RQ6 Computation efficiency (Section V-H): How does the training speed and memory usage of MSGCA comparing to baseline methods?

Experimental setup

1) Datasets: We evaluate the performance of MSGCA on three public benchmark datasets: **BigData22** [15], **ACL18** [7], **CIKM18** [59], and one new dataset that we collect and publish: **InnoStock**. Table 3 presents a summary of these datasets, detailing the count of stocks, the number of news documents (InnoStock) and tweet documents (others), the number of edges in the industry graph, and the duration of trading dates. The three existing datasets consist of high-trade-volume stocks in US stock markets and dynamic input from tweets and price sequences. Innostock, originally collected by us from CSMAR,[3] focuses on newly formed technology companies listed on China's Sci-Tech Innovation Board, aggregating their financial news from various online platforms. To accommodate MSGCA's multimodal input, we further collect the industrial sector relationships for each stock from the above datasets to build graphs. We label each stock according to the increase rate of its adjusted closing prices. Following [15], the increase rate is calculated as $r_t^s = p_t^s / p_{t-1}^s - 1$, where p_t^i is the adjusted closing price of the stock s at the timestamp t. To ensure balanced labels in our fine-grained classification, we adjust the growth rate threshold for flat trends according to each dataset. We set $[-1\%, 1\%]$, $[-0.5\%, 0.5\%]$, $[-0.4\%, 0.4\%]$, and $[-0.3\%, 0.3\%]$ for InnoStock, BigData22, ACL18, and CIKM18, respectively. For example, InnoStock timestamps that have $r_t^s \geq 1\%$ and $r_t^s \leq -1\%$ are labeled as up (1) and down (-1), respectively, and $-1\% < r_t^s < 1\%$ are labeled as flat (0). To ensure a realistic evaluation aligned with time-series prediction requirements, we partitioned each dataset strictly chronologically, consistent with recent studies on stock movement prediction [15, 35]. Specifically, we allocated the first 80% of the time-series data to the training set, the next 10% to the validation set, and the remaining 10% to the testing set. This method prevents data leakage from future timestamps and simulates a practical deployment scenario. No temporal overlap exists among the three splits.

2) Evaluation metrics: We assess the performance of multimodal stock movement prediction using two measures: accuracy (**ACC**) and the Matthews Correlation Coefficient (**MCC**) [60]. ACC is commonly used in various classification issues, while MCC improves the fairness of the evaluation by considering all four confusion matrix indicators: true positives (tp), true negatives (tn), false positives (fp) and false negatives (fn). The MCC can be calculated as follows:

$$MCC = \frac{tp \times tn - fp \times fn}{\sqrt{(tp + fp)(tp + fn)(tn + fp)(tn + fn)}}. \quad (25)$$

3) Hyperparameters: We train our MSGCA and all baseline methods with the following key hyperparameters. The hidden embedding dimensions of all the encoder output and fused stock features are set to 64. A shorter window yielded insufficient temporal context, while longer windows reduced the number of valid samples, limiting model generalization. We fixed the input sequence window size to 20 days based on prior studies [15, 35] and empirical validation. We set the number of heads to 2 for multi-head cross-attention layers. The number of training epochs is 200 for BigData22 and

[3] https://cn.gtadata.com/

Table 3 Summary of important statistics of four datasets

Features	InnoStock (Ours)	BigData22 [15]	ACL18 [7]	CIKM18 [59]
#Stocks	369	50	87	38
#Docs	6,756	272,762	106,271	955,788
Doc Type	news	tweets	tweets	tweets
Market	China SSE STAR	U.S. Market	U.S. Market	U.S. Market
#Edges	385	50	87	26
Date Range	2022-01 to 2022-12	2019-07 to 2020-06	2014-01 to 2015-12	2017-01 to 2017-12

CIKM18 and 250 for InnoStock and ACL18. The batch size is set to 4096 for InnoStock and 1024 for others. The learning rate is 1e-4. We use a warm-up training strategy for all methods with the Adam optimizer. All hyperparameters of the baseline methods are set as the numbers reported in the original papers. We run each method five times with different random seeds and report the average performance and variances.

Baseline methods

We compare the performance of our MSGCA with three categories of baselines for multimodal stock movement prediction as follows:

1) *Indicator-only methods*, which focus only on price input to perform prediction, including **LSTM** [2] and **ALSTM** [61]. LSTM is a representative model for sequential data. ALSTM combines the hidden states of LSTM with the output of attention.

2) *Indicator-document methods*, which combine the features of indicator sequences and dynamic documents to predict future trends, including **ALSTM-W** [15], **SLOT** [15], **LLMFactor** [66] and **FinedFMP** [67]. ALSTM-W computes the average embeddings of documents using a pretrained language model along with indicator features. SLOT integrates price features and two types of trend features with ALSTM. LLMFactor and FinedFMP employ LLMs to integrate financial news for forecasting. We partially implement these LLM-based methods to compare their architectural designs and performance using GPT-4o-mini,[4] and denote it as **LLM-Stock**.

3) *Indicator-graph methods*, that use graph structural information to improve performance, including **ESTIMATE** [26] and **DTML** [35]. ESTIMATE concatenates price features with stock hypergraph features for prediction. DTML uses attentive LSTM to compute features of stocks and market relations and integrates them with concatenation and attention.

4) *Indicator-document-graph methods*, which fuses the three modality features naively via sequential cross-attention without the gating mechanism, using the same encoders and predictor as MSGCA, denoted as **MSGCA-NAF**.

We run the above baseline methods either by modifying their open-source implementations to fit our settings or by reimplementing from the descriptions in their original papers.

Performance comparison (RQ1)

Stock movement prediction performance is evaluated using two aforementioned metrics. The results are shown in Table 4, where the best results are highlighted in bold, and the most competitive baselines are underlined. Our MSGCA consistently achieves the best performance on all datasets and metrics. Specifically, we have the following observations.

1) MSGCA outperforms baselines on all datasets: Due to the introduction of more modalities and the stable fusion modules, our MSGCA achieves the best results on all datasets and metrics for the fine-grained stock movement prediction task. Considering the MCC metric, the improvements in MSGCA compared to the second-best results on four datasets are 8.1%, 6.1%, 21.7% and 31.6%, respectively. Meanwhile, the best competitors on four datasets are SLOT or DTML. These two methods have been proposed in recent years and have more advanced architectures than other baselines, which shows consistency with their results.

2) Modalities contribute differently on each dataset: Given the variability in stocks, time spans, and data origins, the three modalities could have varying impacts on predictive accuracy. Regarding the performance of baseline methods, dynamic documents of stocks have a greater impact on ACL18 and InnoStock compared to their influence on BigData22 and CIKM18. A possible reason is that ACL18 contains more news coverage per stock due to its longer time span, which results in richer text signals, and InnoStock focuses on Chinese Sci-Tech stocks, which are more sensitive to news due to their emerging nature and investor sentiment. On the other hand, industry relationships, which facilitate the propagation of information between stocks, prove to be more effective in BigData22 compared to others. However, the

[4] https://platform.openai.com/docs/models/gpt-4o-mini.

Table 4 Performance comparison of stock movement prediction. The best results are in bold, and the second-best results are underlined. Our proposed MSGCA demonstrates the best performance across all four datasets and two evaluation metrics

Methods	InnoStock (ours)		BigData22 [15]	
	ACC	MCC	ACC	MCC
LSTM [2]	0.3545±0.0005	0.0320±0.0011	0.4119±0.0023	0.0866±0.0059
ALSTM [61]	0.3762±0.0017	0.0339±0.0014	0.4335±0.0079	0.0968±0.0072
DTML [35]	0.4208±0.0009	0.0493±0.0013	0.4367±0.0123	0.1048±0.0131
ESTIMATE [26]	0.3995±0.0020	0.0482±0.0046	0.4296±0.0027	0.1004±0.0059
ALSTM-W [15]	0.3835±0.0015	0.0431±0.0022	0.4287±0.0025	0.0999±0.0066
SLOT [15]	0.4127±0.0006	0.0509±0.0005	0.4250±0.0082	0.0980±0.0081
LLM-Stock [66, 67]	0.4059±0.0002	0.0496±0.0001	0.4215±0.0002	0.0937±0.0001
MSGCA-NAF	0.4052±0.0006	0.0501±0.0010	0.4345±0.0053	0.1029±0.0046
MSGCA	**0.4223±0.0003**	**0.0550±0.0008**	**0.4379±0.0077**	**0.1112±0.0037**
Methods	ACL18 [7]		CIKM18 [59]	
	ACC	MCC	ACC	MCC
LSTM [2]	0.3779±0.0022	0.0254±0.0020	0.3781±0.0006	0.0393±0.0028
ALSTM [61]	0.3806±0.0027	0.0275±0.0031	0.3830±0.0011	0.0470±0.0046
DTML [35]	0.3832±0.0053	0.0306±0.0085	0.3811±0.0080	0.0613±0.0072
ESTIMATE [26]	0.3805±0.0046	0.0312±0.0074	0.3806±0.0047	0.0579±0.0035
ALSTM-W [15]	0.3848±0.0050	0.0375±0.0064	0.3818±0.0082	0.0545±0.0096
SLOT [15]	0.3894±0.0016	0.0487±0.0168	0.3848±0.0011	0.0601±0.0053
LLM-Stock [66, 67]	0.3795±0.0001	0.0319±0.0002	0.3798±0.0001	0.0549±0.0002
MSGCA-NAF	0.3816±0.0032	0.0481±0.0125	0.3826±0.0062	0.0769±0.0085
MSGCA	**0.3966±0.0050**	**0.0593±0.0073**	**0.3861±0.0023**	**0.0807±0.0045**

trimodal encoder and gated cross-attention of our MSGCA efficiently integrate information across all three modalities, enhancing overall performance.

3) Introducing more modalities may damage accuracy: Inconsistent multimodal datasets cause unstable feature fusion in baseline methods, leading to inaccurate predictions. Specifically, when integrating dynamic document modalities, we observe a relative decline in accuracy on BigData22, ACL18, and CIKM18 datasets compared to the unimodal ALSTM frameworks. However, our MSGCA effectively processes noisy document data and achieves accurate predictions, leveraging a gated cross-attention mechanism for reliable integration. Additionally, our InnoStock dataset reveals a consistent performance improvement when integrating news documents and relational graphs, demonstrating the superior quality of our proposed datasets.

Ablation Study (RQ2)

In this subsection, we examine how various components within the encoding and fusion modules impact MSGCA's performance. We denote the MSGCA variants as "MSGCA-X", which represents the framework to replace a specific part of it with "X". Our study focuses on three aspects as follows.

1) Impact of different fusion strategies: We evaluate the impact of fusion strategies by replacing gated cross-attention blocks in MSGCA with two foundational networks: gated layer unit (GLU) [52] and cross-attention [46]. These two variants are denoted as **MSGCA-GLU** and **MSGCA-CA**, respectively. From the performance comparison of the three frameworks shown in Fig. 3, the GLU fusion approach is direct yet overly simplistic to effectively manage the interactions between multimodal features for competitive predictions. The cross-attention mechanism can effectively aggregate features from multiple modalities, but suffers from noisy information and leads to performance decline. Furthermore, the results on BigData22 indicate a more detrimental fusion between modalities compared to other datasets.

2) Impact of language encoding models: A language encoding model transforms document text into latent embeddings for the subsequent fusion process. In our study, we evaluated three pre-trained language models, including BERT [63], ModernBERT [68], and OpenAI's text-embedding-ada-002 to preprocess and manage dynamic document modality input. These language models vary in parameter size and output dimensionality. The frameworks using these three language models are denoted as **MSGCA-BERT**, **MSGCA-MBERT**, and **MSGCA**, respectively. The results shown in Fig. 4 illustrate that prediction performance improves along with the size and the power of the language model employed. Furthermore, the advanced large language model (Ada) used

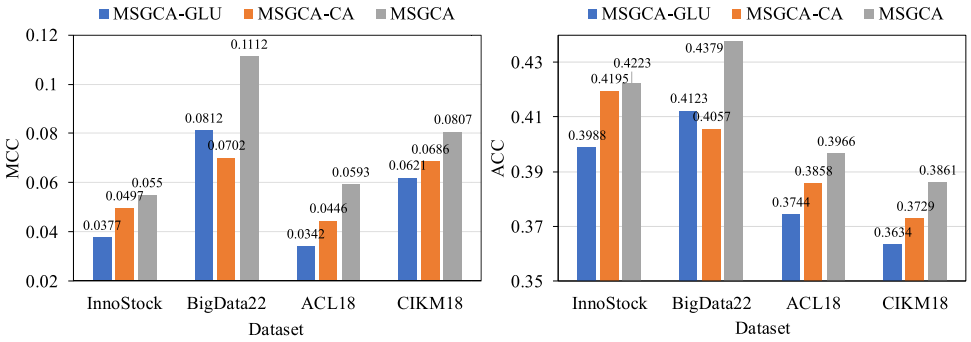


Fig. 3 Performance impacts of different fusion strategies on four datasets. MSGCA-GLU cannot manage the interaction between multimodal features, while MSGCA-CA suffers from noisy information aggregation

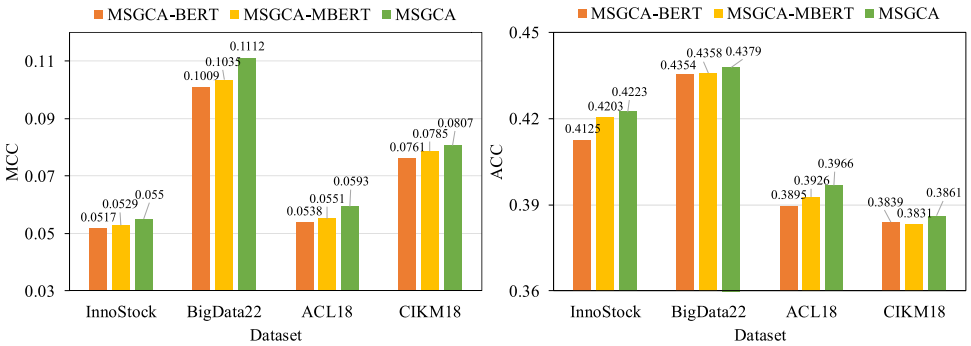


Fig. 4 Impacts of different language encoding models on four datasets for handling dynamic documents. Embeddings generated from the advanced large language model can significantly improve the performance

into our final version of MSGCA outperforms the two smaller language models, delivering superior results.

3) Impact of graph encoding models: To evaluate the impact of graph encoding models, we extract structural insights from stock interrelations by applying multiple graph encoding techniques, such as Random Walk [64], GCN [65], and GAT [56]. These models aggregate information from neighboring nodes with different selection and computation strategies to obtain central node features. We denote frameworks using these three graph encoders as **MSGCA-RW**, **MSGCA-GCN**, and **MSGCA**. As shown in Fig. 5, models considering a higher order of relationships can achieve better performance. Our MSGCA, which employs GAT, can implicitly specify different weights to all nodes in a neighborhood while encoding graph structures and finally gains the best results compared to other graph encoding models.

Effect of multimodal sources (RQ3)

In this subsection, we examine how various modalities aid in predicting stock movement. To evaluate this, we implement three variants of MSGCA where a modality is removed from our trimodal encoder module: (1) **MSGCA-ID**: a version in which indicator sequences and dynamic documents are considered without a relational graph as input; (2) **MSGCA-IG**: a version that indicator sequences and graph are kept for feature fusion; (3) **MSGCA-DG**: a version using only documents and relationships, excluding the primary indicator modality. The performance of these variants on four datasets is shown in Table 5. We can observe from the results that MSGCA-ID and MSGCA-IG still outperform existing stock movement prediction methods on most metrics, demonstrating the capability of our feature fusion and movement prediction modules. For example, the MCC of MSGCA-ID is 1.7% and 9.8% higher than that of SLOT on InnoStock and BigData22, respectively. In addition, the modality of indicator sequences plays a crucial role in the prediction performance. All results of MSGCA-DG decrease dramatically after the indicator modality is removed from those of other frameworks. Furthermore, without relying on indicator features, MSGCA can yield results compared to baseline methods that use indicators alone, showing its capability to effectively utilize the information from graph relationships and the textual data of news and tweets for prediction. Furthermore, removing dynamic documents reduces performance more significantly on ACL18 and InnoStock, which shows consistency with the observations from the results of the performance comparison in Section 5.3.

Study for stable fusion (RQ4)

To demonstrate that our MSGCA is able to perform a stable feature fusion for multimodal stock movement prediction, we further analyze the reliability of MSGCA during the progressive fusion phase across three modalities. We randomly select a representative stock from the InnoStock dataset. Embedding vectors of the stock, derived from the fused features, are sequentially dumped to files following each of the two fusion stages. Then, we reduce the feature dimensions into one-dimensional vectors via Principal Component

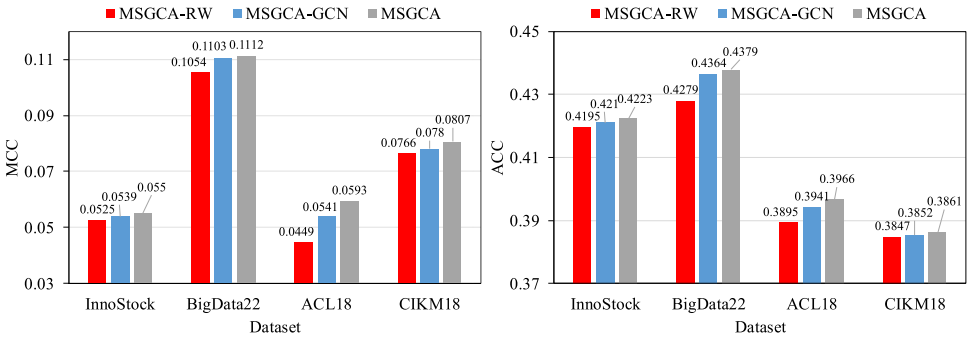


Fig. 5 Impacts of different graph encoding models on four datasets for handling graph structural information. Graph attention network enables nodes to attend over their neighboring features and helps the most for the overall performance

Table 5 Impacts of three modalities on four datasets. Performance degradation percentages are shown in Δ. The best results are in bold

Methods	InnoStock (ours)				BigData22 [15]			
	ACC	Δ	MCC	Δ	ACC	Δ	MCC	Δ
MSGCA-ID	0.4163	1.4%	0.0518	5.8%	0.4309	1.6%	0.1076	3.2%
MSGCA-IG	0.4059	3.9%	0.0525	4.5%	0.4273	2.4%	0.1006	9.5%
MSGCA-DG	0.3569	15.5%	0.0344	37.5%	0.4052	7.5%	0.0775	30.3%
MSGCA	**0.4223**	-	**0.0550**	-	**0.4379**	-	**0.1112**	-
Methods	ACL18 [7]				CIKM18 [59]			
	ACC	Δ	MCC	Δ	ACC	Δ	MCC	Δ
MSGCA-ID	0.3860	2.7%	0.0354	41.8%	0.3811	1.3%	0.0664	17.7%
MSGCA-IG	0.3889	1.9%	0.0499	15.9%	0.3835	0.7%	0.0738	8.6%
MSGCA-DG	0.3591	9.5%	0.0189	68.1%	0.3778	2.1%	0.0401	50.3%
MSGCA	**0.3966**	-	**0.0593**	-	**0.3861**	-	**0.0807**	-

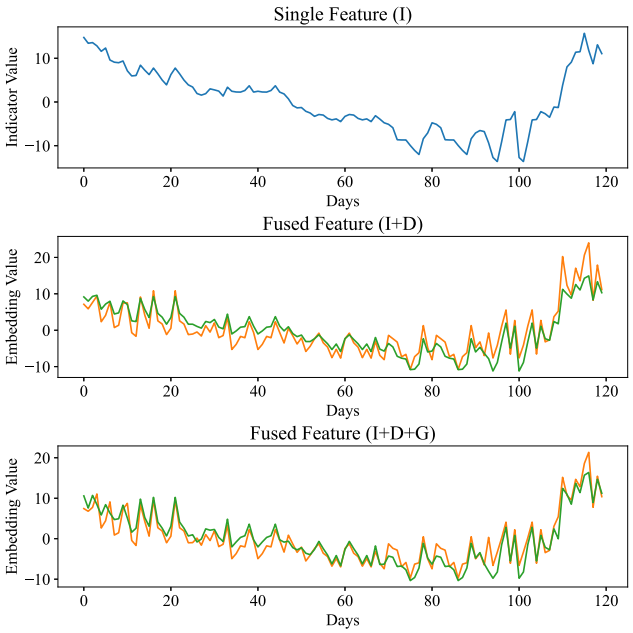


Fig. 6 Stable fusion study by reducing embedding dimensions of a stock to a single dimension, allowing for direct comparison with its price trend. The blue line denotes the price time-series of the stock. The yellow lines represent features after cross-attention fusion without guidance by the primary modality (indicator sequences), which exhibit greater volatility. The green lines indicate features obtained through gated cross-attention blocks guided by indicator features, showing smoother transitions and stronger alignment with the price trend

Analysis (PCA), which are then aligned with the stock's closing price timeline for comparative analysis across various fusion stages. This comparison is shown in Fig. 6. Visualizing the latent features of InnoStock reveals that the gated layers of MSGCA, coupled with a cross-attention mechanism, can mitigate issues of data sparsity and semantic conflicts. To be more specific, two green lines, produced by gated cross-attention blocks, are smoother and more aligned with the price trajectory compared to the yellow lines, which are derived from feature fusion without primary modality guidance and display significant volatility. Meanwhile, the smoothing effect takes place after each fusion stage (I+D and I+D+G), indicating that our gated cross-attention fusion strategy consistently addresses the data challenge across multiple modalities.

Effect of hyperparameters (RQ5)

1) Impact of hidden dimension size: We investigate the impact of different dimension sizes of hidden embeddings in MSGCA, including the representations of each modality and the features after fusion. We set the dimension size $d \in [16, 32, 64, 128]$. The MCC and ACC variation curves for the four datasets are shown in Fig. 7. We can observe that the performance of MSGCA increases along with the sizes from 16 to 64, and decreases with the size of 128 on all

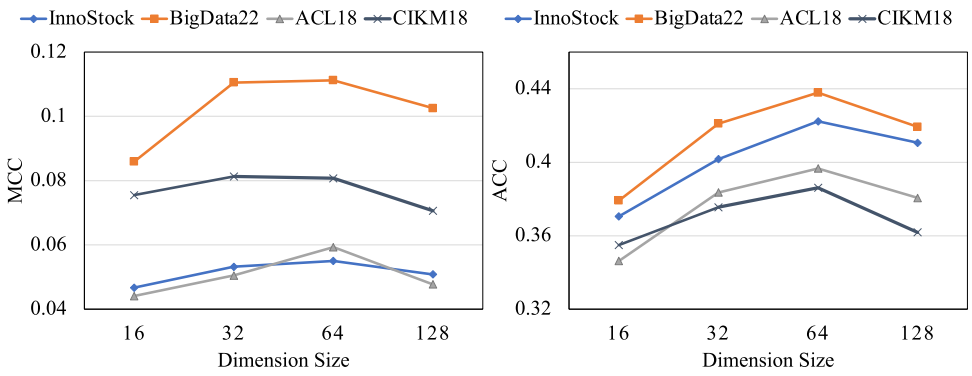


Fig. 7 The MCC of MSGCA with various dimension sizes of hidden embeddings on four datasets

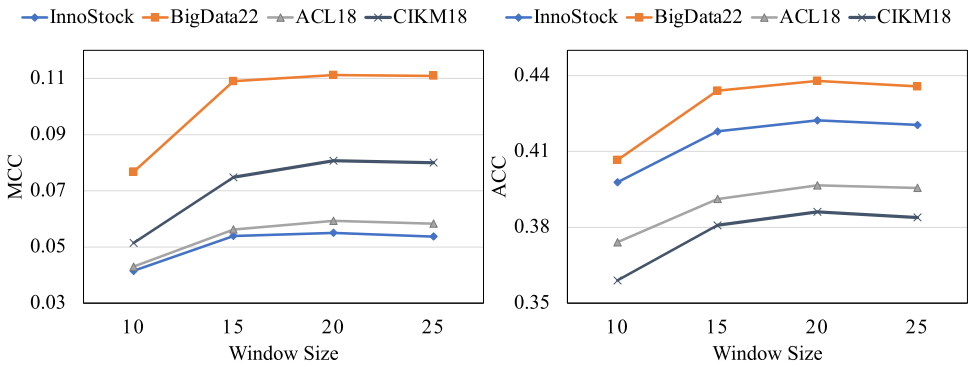


Fig. 8 The performance of MSGCA with various window sizes for input sequences on four datasets

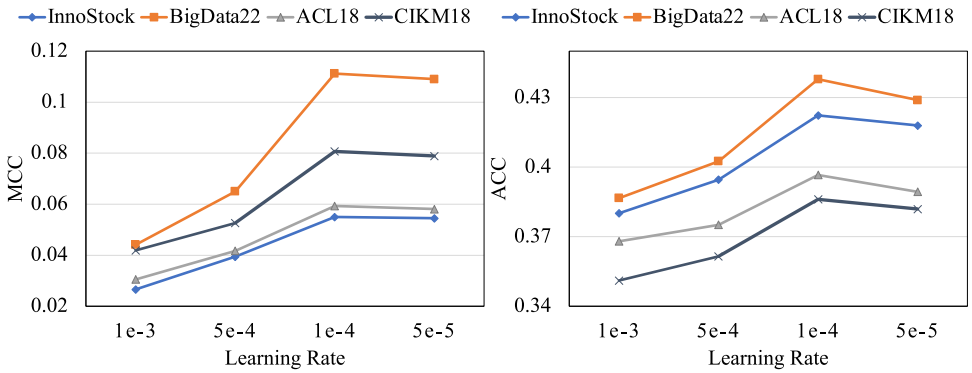


Fig. 9 Performance of MSGCA with various learning rates on four datasets

datasets. This illustrates a lack of expression with smaller dimensions and an overfitting with larger dimensions. In addition, ACL18 and BigData22 show more sensitive to hidden dimension size than the other two datasets. Considering both the performance and efficiency, we choose $d = 64$ as the best hidden dimension size.

2) Impact of time window size: We evaluate the influence of window size in MSGCA. Following settings from previous studies [15, 35], we set the window size $ws \in [10, 15, 20, 25]$ which balances the length of sequences and the amount of training samples. The MCC and ACC results on four datasets with different window sizes are shown in Fig. 8. The performance of MSGCA dramatically decreases with window size of 10 due to a lack of useful information from the time series. Meanwhile, MSGCA cannot gain further improvement from larger window sizes because of fewer training samples obtained from datasets, which limits the information to be learned. In our experiments with the best results, we set $ws = 20$ as the window size.

3) Impact of learning rate: Training strategies significantly affect the performance of MSGCA, and the learning rate is a crucial training parameter. As shown in Fig. 9, a too slow gradient descent (with the learning rate set to $5e-5$) can hardly reach optimal results within reasonable training steps, leading to a low-efficiency training process. Meanwhile, a training speed that is too fast (with the learning rate set to

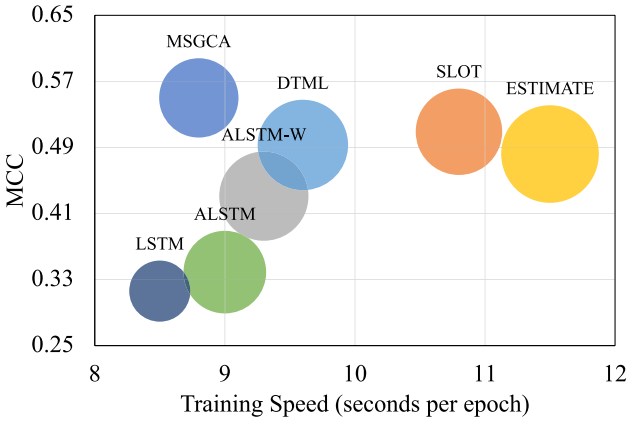


Fig. 10 Computational efficiency comparison on InnoStock dataset. MCC performance (y axis), training speed (x axis), and memory usage (size of the circles) of methods are shown

$1e-3$ and $5e-4$) may skip optimal results and generate sub-optimal performance. With a large number of experiments, we set the learning rate $lr = 1e-4$ as the best in MSGCA.

Computational efficiency analysis (RQ6)

In this subsection, we analyze the computational efficiency of MSGCA compared with baseline methods in two aspects: training speed and the memory usage. We evaluate all methods with the same batch size and record the average time usage per epoch for each method. On the other hand, we monitor the CUDA memory usage by dumping snapshots of memory allocations during training and record the largest memory value for each method. The comparison of computation efficiency on InnoStock dataset is shown in Fig. 10. On one hand, our MSGCA delivers superior results with a faster training process and less memory usage than current leading stock movement prediction methods. On the other hand, MSGCA achieves cost-effective computation while outperforming methods with simpler architecture.

Conclusion and future work

In our study, we examine the challenges of multimodal stock movement prediction using a stable fusion approach via a gated cross-attention mechanism. Our contribution, MSGCA, addresses the task by leveraging heterogeneous data sources and fills the gap with the study of the fine-grained stock movement prediction task with multimodal information. The MSGCA framework begins with a trimodal encoder to handle three distinct data modalities: indicator sequences, dynamic documents, and a relational graph. This is followed by a stable fusion module to subsequently integrate these three modalities with a pair of gated cross-attention networks guided by primary features. The final phase involves deploying a movement prediction module consisting of two MLP networks to map the integrated features into predictive signals to perform the fine-grained prediction task. Extensive experiments and analysis on four multimodal stock datasets demonstrate the effectiveness and efficiency of MSGCA. In future work, we aim to enrich our model with more features from various modalities and explore its fusion capabilities in different application domains.

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Author Contributions Hang Zhou provided overall guidance and supervision for the project. Chang Zong led the development and implementation of the trimodal encoding module and the cross-feature fusion module. Jian Wan contributed significantly to the design and optimization of the gated cross-attention mechanisms. Lucia Cascone provided critical theoretical insights into the fusion stability and overall model architecture.

Data availability Availability of data and code Source code for MSGCA and four datasets are all published and available from our GitHub repository at https://github.com/changzong/MSGCA.

Declarations

Conflict of interest The authors have no Conflict of interest to declare that are relevant to the content of this article.

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